

RVP — Running Experiment Log (all results, honest)

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All metrics are single-attempt pass@1 (unbiased from $k=16$) unless noted; coverage held $\approx$ constant. Controls per cell: base, RFT, RVP (ours), *xrft* (positive-only), *shuf* (shuffled-pair). Nulls and ties are reported as findings. Ledger § tags reference ADAPTIVE_FORGETTING_RESULTS.md.

1 Generalization: final-24 matrix (§180) — 3 models $\times$ 6 datasets $\times$ 4 seeds

Seed-averaged $\Delta\text{pass@1} = \text{rvp} - \text{base}$; ceiling cells † (base $> .85$).

Dataset	Math-1.5B (base)	Math-1.5B-Inst	Math-7B-Inst
GSM8K	+.206 (.508 $\rightarrow$.714)	−.004 †	+.002 †
MATH-500	+.100 (.485 $\rightarrow$.585)	+.005	+.010
AMC	+.089 (.281 $\rightarrow$.370)	+.025	+.016
DeepMath	+.074 (.282 $\rightarrow$.356)	−.003	+.014
Olympiad	+.039 (.235 $\rightarrow$.273)	+.002	+.025
Omni-MATH	+.035 (.148 $\rightarrow$.183)	+.011	+.014

Verdict: math-specialized *base* wins all 6 (+.04 to +.21, per-seed spread $\leq .006$); both *instruct* variants near-flat (gap consumed by instruction-tuning). Headroom-gated scope boundary.

2 7B full-param sharded headline (§170) + generalization across sizes

Qwen2.5-Math-7B (sharded ZeRO-3 DPO), RVP top arm all 3: MATH-500 .597 $\rightarrow$.734 (**+.137**), Olympiad .295 $\rightarrow$.353 (+.058), Omni .197 $\rightarrow$.252 (+.055). Qwen2.5-Math-1.5B: +.172/.089/.046 (3-seed). *General* base (Qwen2.5-7B) mixed; RLHF-chat (Yi-1.5-9B-Chat) collapses (−.125 to −.164). Discriminator: axis is **math-specialized pretraining**, not instruct-tuning (Qwen2.5-Math-7B-Instruct wins all 3).

3 Reliability ladder (§166–167): base < GRPO $\approx$ base < RFT < RVP

Direct GRPO arms (finite-budget): GSM8K Qwen-1.5B base .653 / GRPO .673 / RFT .711 / **RVP .747**; deepseek-1.3B .026/.028/.050/.**072**. GRPO stays near base at pass@1 (Cor. 1.1, a budget/signal statement, not impossibility).

4 Mechanism (§163/167/180): margin ascent via incorrect-mode suppression

Teacher-forced $\log \pi$ on the SAME held-out verified pairs (GSM8K Qwen-1.5B):

stage	$\log \pi(y^+)$	$\log \pi(y^-)$	margin m
base	−.110	−.139	.029
GRPO	−.109	−.139	.029 (=base)
RFT	−.092	−.119	.027 (both up; flat)
RVP	−.092	−.157	.065 (suppress y^- ; 2.4 $\times$)

shuf	−.091	−.119	.027 (=RFT)
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7B margin rises 25–30×. Layer-wise logit-lens: margin-gain concentrates in **late layers** (26/28 for 1.5B, 21/28 for 7B) — RVP sharpens the final decision, not early features.

5 Reliability–compute frontier (§181): RVP amortizes inference-time selection

RVP@1 (one forward pass) vs base majority-vote self-consistency@ n . n^* = samples base needs to reach RVP@1.

Dataset	base@1	RVP@1	base maj@8	n^*
GSM8K	.517	.718	.723	~8
MATH-500	.487	.591	.629	~8
AMC	.280	.362	.433	~8
DeepMath	.279	.364	.361	~16
Olympiad	.235	.273	.326	~4
Omni-MATH	.149	.182	.184	~8

RVP@1 $\approx$ base self-consistency at ~ 8 samples (median) — an $\sim 8\times$ inference budget baked into the weights; stays below oracle best-of- n (a realistic verifier-free single-shot selector).

6 Iterative-RFT (ReST) baseline + mechanistic dissociation (§181/183)

pass@1 (base Qwen2.5-Math-1.5B): RVP beats base+RFT on all cells; RVP-vs-ReST is a *near-tie* (ReST edges GSM8K, RVP edges Olympiad/AMC). Single-seed, 150–200 problems: treated as statistically equivalent pending a powered test (± 1 pt margin $\Rightarrow \sim 1.8$ k problems, ≥ 3 seeds). **Probability-mass accounting** (aggregate margin $M = \log \sum_{\mathcal{C}} \pi - \log \sum_{\mathcal{I}} \pi$, correct-mass $s = \sigma(M)$, pooled support):

cell	base M/s	RFT M/s	RVP M/s	ReST M/s
GSM8K	20.95/.788	19.87/.801	21.65/.818	19.59/.808
Olympiad	20.82/.759	20.36/.750	22.16/.768	19.92/.747
AMC	20.41/.697	19.32/.701	21.25/.728	18.96/.702

Dissociation: RVP has the highest aggregate margin on every cell; ReST matches/beats pass@1 but does *not* widen the aggregate margin (at/below RFT). Same pass@1 endpoint, different internal route — RVP via genuine selection, ReST without it. Rescues the mechanism (RVP-specific) while conceding the pass@1 tie. Instruct-model mass (§182): RVP raises aggregate margin 2–4× on all 6 cells even though pass@1 is flat $\Rightarrow$ aggregate-margin $\uparrow$ is *necessary but not sufficient* (needs decode-time selection headroom the instruct model lacks) — deepens the mechanism and explains the instruct-flat result.

7 Ablations (§171/177)

Data efficiency: ~ 25 verified pairs capture the full gain ($\Delta + .108@25$ vs $+ .112@800$). **One-shot:** iterating RVP collapses it ($-.118$ MATH-500, $-.154$ Omni) — single-pass is optimal. **Hard-neg mining:** statistical tie with random verified negatives ($\Delta \leq \pm .006$). **β -robustness:** net-positive for $\beta \in \{.1, .2, .5\}$ all datasets; only $\beta=.03$ over-optimizes. **RVP-from-base:** reaches only .24–.28 vs .62–.65 from RFT $\Rightarrow$ the RFT coverage stage is necessary.

8 Scope boundaries + honest retractions

Wins: math-specialized base *and* instruct. Boundaries: general base mixed; RLHF-chat (Yi) collapses; Mathstral collapses because RFT created no gap (precondition = a successful RFT gap). Retractions: §169 withdrew a confounded RVP-from-base null (over-optimized DPO); §176 corrected the hard-neg “HURTS” artifact to a tie. `competition_math` loader dead upstream $\Rightarrow$ dropped, not guessed.

9 Theory corrections (Astra review, §182)

Prop. 1 rescoped: positive-only CE *does* lower $\log \pi(y^-)$ diffusely via the softmax normaliser ($\partial \log \pi(y^+)/\partial z_{y^-} = -\pi(y^-)$); the honest claim is *untargeted* suppression, empirically dominated by the shared-feature lift $\Rightarrow$ net margin flat (scoped-empirical, not impossibility). Cor. 1.1 softened to a finite-budget/signal statement (GRPO $\approx$ base empirically, not a theorem that outcome-RL can never acquire hard prompts). Contribution reframed: large headroom-gated selection gain + amortization + *predictive* scope, not “verified negatives uniquely necessary.”

10 Config note + limitations (honest)

Two configs (why 1.5B MATH-500 differs across tables): full-param sharded DPO gives +.172 (3-seed); single-GPU LoRA-DPO gives +.125 / +.100 (4-seed final-24) — LoRA updates a low-rank slice, a few points lower; not seed noise. 7B headline is sharded. **Limitations:** (i) headroom-gated — near-flat on instruct, regresses on RLHF-chat (Yi) and where RFT opens no gap (Mathstral); (ii) not uniquely necessary for pass@1 (ReST competitive) but uniquely widens the aggregate margin (dissociation, §183); (iii) several head-to-heads single-seed / $n=150-200$ (powered multi-seed run in progress); (iv) compute accounting is step-count not tokens/GPU-h; (v) verified evidence is math + CompDAG (external code pending); (vi) verifier-free RVP untested.

Pending (2026-09-18, 72 GPUs live)

Multi-seed base-model CIs (bx_m15 seeds 2-4+, A-new/C-new) for powered stats; full 6-cell mass table (m500/om/dm ReST+mass on B). Queued: on-policy/replay bridge (tuned GRPO-from-RFT), verifier surgery (correctness vs format), instruct-tuning trajectory (0/25/100%), external code benchmark.